10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Exercise 9.2 — Circle Angles, Arc Length and Area of Sector

Unit 9 | Solved Exercise | New Book (PECTAA)

Exercise 9.2 (Solved)

Angle at the centre = 2 × angle at the circumference. For arcs and sectors, first change the angle into radians: radians = degrees × π ÷ 180. Then use arc = rθ and area = ½ r²θ.

1. A, B, C and D are points on the circumference of a circle with centre at O as shown in figure. Find the angle x.

In the figure the tangent at C makes 114° with the chord CB, and x is the central angle ∠BOC.

Tangent–chord angle = angle in the alternate segment

∠CDB = 180° − 114° = 66°

Angle at the centre = 2 × angle at the circumference

x = 2 × ∠CDB = 2 × 66°

x = 132°

Answer: x = 132°

2. In the adjoining figure m∠BDC = 74°, find m∠BAC, m∠BOC and m∠OBC.

∠BAC and ∠BDC stand on the same arc BC

∠BAC = ∠BDC = 74°

∠BOC = 2 × ∠BDC = 2 × 74° = 148°

In the isosceles triangle OBC, OB = OC (radii)

$\angle OBC = \angle OCB = \frac{180° - 148°}{2} = \frac{32°}{2} = 16°$

Answer: ∠BAC = 74°, ∠BOC = 148°, ∠OBC = 16°

3. Find the angles x and y in the following figures.

(i) A cyclic quadrilateral ABCD has ∠ABC = 159° and ∠CAD = 74°. (ii) In the circle ∠QPR = 38°, x = ∠PQR and y = ∠PRQ. (iii) Two tangents from X touch the circle at Y and at a second point, and ∠XOY = 62°.

(i) cyclic quadrilateral ABCD

In figure (i) ABCD is a cyclic quadrilateral with ∠ABC = 159° and ∠DAC = 74°.

Opposite angles of a cyclic quadrilateral are supplementary

x = ∠ADC = 180° − ∠ABC = 180° − 159° = 21°

In triangle ADC: y = 180° − 74° − 21°

y = 85°

Answer: x = 21°, y = 85°

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(ii) PR is a diameter

In figure (ii) P, Q, R lie on the circle, PR passes through the centre O and ∠QPR = 38°.

PR is a diameter, so the angle in the semi-circle is 90°

x = ∠PQR = 90°

In triangle PQR: y = 180° − 90° − 38°

y = 52°

Answer: x = 90°, y = 52°

(iii) two tangents from X

In figure (iii) XY and the second line from X are tangents to the circle and ∠XOY = 62°.

The radius OY is perpendicular to the tangent XY

x = ∠OYX = 90°

In the right triangle OXY: y = 180° − 90° − 62°

y = 28°

Answer: x = 90°, y = 28°

4. Find the angle x in the given figure.

In the figure AB and AD are equal chords, AC passes through the centre O and the tangent at C makes 70° with the chord CB.

Tangent–chord angle = angle in the alternate segment

∠BAC = 70°

AC is a diameter, so ∠ABC = 90°

∠ACB = 180° − 90° − 70° = 20°

Equal chords AB and AD subtend equal angles at C

x = ∠ACD = ∠ACB = 20°

Answer: x = 20°

5. In a circular arch over a monument entrance, two spotlight fixtures are placed such that they each shine from two different points on the arch to the same chord PQ on the base of the arch. If the angle formed at one light is 55°, what is the angle at the second light on the same side of chord PQ?

Both lights stand on the same side of the chord PQ

Angles in the same segment of a circle are equal

Answer: The angle at the second light is also 55°

6. A circular garden has a walking path forming a quadrilateral inscribed in it. If one angle is 87°, what is the opposite angle?

Opposite angles of a cyclic quadrilateral are supplementary

Opposite angle = 180° − 87°

= 93°

Answer: The opposite angle = 93°

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7. A ferris wheel has a radius of 12 m. If a passenger travels a distance of 18 m along the circumference of the ferris wheel, what is the angle (in radians) swept by the passenger's position from the starting point?

arc = rθ, so θ = arc ÷ r

$\theta = \frac{18}{12}$

θ = 1.5 radian

Answer: Angle swept = 1.5 radians

8. Find θ, when (i) l = 3 cm, r = 2.2 cm (ii) l = 5.6 cm, r = 2 cm

(i) l = 3 cm, r = 2.2 cm

$\theta = \frac{l}{r} = \frac{3}{2.2}$

θ = 1.364 radian

Answer: θ = 1.36 radian

(ii) l = 5.6 cm, r = 2 cm

$\theta = \frac{5.6}{2} = 2.8$

Answer: θ = 2.8 radians

9. Find r, when (i) l = 5.5 cm, θ = 40°20′ (ii) l = 13 cm, θ = 70°

(i) l = 5.5 cm, θ = 40°20′

$\theta = 40°20' = 40.333° = 40.333 \times \frac{\pi}{180} = 0.704$ radian

$r = \frac{l}{\theta} = \frac{5.5}{0.704}$

r = 7.81 cm

Answer: r = 7.81 cm

(ii) l = 13 cm, θ = 70°

$\theta = 70 \times \frac{\pi}{180} = 1.222$ radian

$r = \frac{13}{1.222} = 10.64$ cm

Answer: r = 10.64 cm

10. Find l and area of sector, when (i) r = 1.7 cm, θ = 0.25 radian (ii) r = 3 cm, θ = 45°

(i) r = 1.7 cm, θ = 0.25 radian

l = rθ = 1.7 × 0.25 = 0.425 cm

$\text{Area} = \frac{1}{2} r^2\theta = \frac{1}{2} \times (1.7)^2 \times 0.25$

$= \frac{1}{2} \times 2.89 \times 0.25 = 0.361$ cm²

Answer: l = 0.425 cm, area = 0.361 cm²

(ii) r = 3 cm, θ = 45°

$\theta = 45 \times \frac{\pi}{180} = 0.7854$ radian

l = rθ = 3 × 0.7854 = 2.36 cm

$\text{Area} = \frac{1}{2} \times 3^2 \times 0.7854 = 3.53$ cm²

Answer: l = 2.36 cm, area = 3.53 cm²

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11. Uzma cut a pizza of radius 14 cm into 8 equal slices. What is the area of one slice (sector).

Angle of one slice = $\frac{360°}{8}$ = 45° = 0.7854 radian

Area = $\frac{1}{2}$ $r^2\theta$ = $\frac{1}{2}$ × 14^2 × 0.7854

= $\frac{1}{2}$ × 196 × 0.7854

= 76.97 cm²

Answer: Area of one slice ≈ 76.97 cm²

12. The perimeter and area of a sector are 14 cm and 10 cm² respectively. Find the radius of the circle and the central angle of the sector.

Perimeter of sector = 2r + l = 14 ..... (i)

Area of sector = $\frac{1}{2}$ rl = 10 ..... (ii)

From (ii): rl = 20 $\Longrightarrow$ l = $\frac{20}{r}$

Put in (i): 2r + $\frac{20}{r}$ = 14

$2r^2 - 14r + 20 = 0$

$r^2 - 7r + 10 = 0$

(r − 5)(r − 2) = 0

r = 5 cm or r = 2 cm

For r = 5: l = $\frac{20}{5}$ = 4, θ = $\frac{l}{r}$ = $\frac{4}{5}$ = 0.8 radian

For r = 2: l = 10, θ = 5 radians (reflex, so rejected)

Answer: r = 5 cm, θ = 0.8 radian

Remember — circle angles and sectors

Angle at the centre = 2 × angle at the circumference.

Angles in the same segment are equal; angle in a semi-circle is 90°.

Opposite angles of a cyclic quadrilateral add to 180°.

Tangent–chord angle = angle in the alternate segment.

Radians = degrees × π ÷ 180; arc = rθ; area of sector = ½ $r^2\theta$.